# Atomic-Unit Decomposition and Relation Inference

Worked Examples — Master Index

FactReasoner — Ideation

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## Overview

This collection demonstrates a single recurring task applied to several kinds of text: decompose a paragraph into *self-contained atomic units*, then infer two relation types between them —

- **logical prerequisite** ($A \to B$): $A$ must be true for $B$ to be true or meaningful;
- **position invalidation / contradiction** ($A \neq B$): $A$, true at its position in the text, contradicts a claim $B$ appearing later (or a later atom overrides an earlier one).

Each example is a standalone document with its source paragraph, atomic units, relation lists, and a TikZ relation graph. Click a title to open that example's PDF (build the examples first; see the header of `index.tex`).

## Examples

1. **1. A legal damages paragraph**
   `example-1-damages.tex` → `example-1-damages.pdf`
   A contract-damages paragraph in which a mid-text ruling that the agreement was *void ab initio* retroactively invalidates the earlier breach-of-contract claims, while a separate unjust-enrichment award survives. Introduces both relation types and the pivotal-invalidation pattern.

2. **2. A biography (with planted contradictions)**
   `example-2-biography.tex` → `example-2-biography.pdf`
   The Lanny Flaherty biography: first as an internally consistent passage (pure prerequisite / subsumption structure, no contradictions), then a modified version with several planted contradictions (birth year, birthplace, nationality, career start, voice) drawn as pairwise invalidation edges.

3. **3. A narrative passage (Elinor)**
   `example-3-narrative.tex` → `example-3-narrative.pdf`
   A scene from *Sense and Sensibility* with a late revelation (Robert, not Edward, married Lucy) that invalidates the passage's opening premise. Includes a ground-truth re-decomposition, an alignment table, and a temporal-order discrepancy plot comparing narration order to chronology.

4. **4. Synthesizing a summary from a reliable and an unreliable source**
   `example-4-summary.tex` → `example-4-summary.pdf`
   A summary woven from an accurate source A and a sensational, error-laden source B (the

Keyara Jones example), engineered so later reliable atoms positionally invalidate earlier false ones, including a self-contradictory statistic.

5. **5. A real legal case (*R v Renda*)**
   `example-5-renda.tex` → `example-5-renda.pdf`
   The lead appeal in *Renda, R v* [2005] EWCA Crim 2826. Two summaries of the same appeal: an engineered ordering (S) that manufactures positional contradictions, and a faithful, naturally-ordered summary (K) whose relations collapse to a pure implication chain — illustrating that positional invalidation depends on how a text is arranged, not only on its facts.

## Relation-graph conventions

Across the examples the graphs share a common visual language: solid blue arrows are logical prerequisites ($A \to B$); dashed red arrows are contradictions / position invalidations ($A \neq B$); pivotal or holding/revelation nodes are highlighted (orange), false or invalidated atoms are tinted red, and reliable or surviving atoms are tinted green.